Appendix
ONNX and TOSA based Portability and Rapidus / Tenstorrent Inference Architecture

Prepared by Akira on Dec. 04, 2025

The proposed architecture separates model provenance from the inference hardware ecosystem. Models may originate from NVIDIA based training, AWS Trainium, or other training environments. Regardless of their training origin, a model can enter the proposed inference pipeline when its computational graph is representable within a defined standard ONNX operator subset.

The model is first represented using the standard ONNX operator subset and then lowered to TOSA. In this architecture, ONNX serves as the model level interchange representation, while TOSA serves as the compiler level intermediate representation. TOSA does not eliminate the need for hardware specific optimization; rather, it provides a common computational representation from which target specific optimization and lowering can be performed.

The resulting pipeline is:
NVIDIA / Trainium: AWS / other training platform
→ standard ONNX operator subset
→ TOSA
→ hardware specific optimization and lowering
→ Rapidus/Tenstorrent inference architecture

This architecture therefore does not require an NVIDIA trained model to remain dependent on NVIDIA hardware or software during inference. If the model can be expressed within the supported ONNX operator subset, it can be lowered through TOSA and subsequently optimized for the Rapidus/Tenstorrent inference architecture.

The same principle applies to models trained on AWS Trainium. Trainium can serve as the designated training platform without making Trainium the mandatory inference platform. The training environment and the inference environment are consequently treated as separate architectural decisions.

TOSA provides an additional layer of abstraction between the standardized model representation and the physical inference hardware. It does not, however, eliminate hardware specific compilation or optimization. After conversion to TOSA, the computational graph still has to be optimized and lowered according to the characteristics of the target hardware and its execution environment. In the proposed architecture, this final stage is directed toward the Rapidus/Tenstorrent inference architecture.

The NVIDIA trained model case is particularly important as a practical test of the proposed separation between model provenance and inference hardware. If a model originally trained on NVIDIA hardware can be represented within the supported ONNX operator subset, lowered to TOSA, and subsequently optimized for Rapidus/Tenstorrent inference, then the inference deployment is no longer intrinsically determined by the hardware ecosystem in which the model was trained.

The objective is not unrestricted hardware portability. Instead, the architecture provides controlled portability at the model and compiler representation layers while retaining hardware specific optimization at the final stage. ONNX provides the model level interchange boundary, TOSA provides the compiler level abstraction, and the hardware specific backend layer performs the optimization and lowering required for efficient edge inference.

This layered architecture allows models originating from different training ecosystems, including NVIDIA, to be redirected toward the Rapidus/Tenstorrent inference architecture without requiring the inference environment to reproduce the original training hardware and software stack.

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Note-1: Models containing unsupported operators would require additional conversion or adaptation before entering the proposed pipeline.

Note-2: Portability of the computational representation does not by itself guarantee equivalent performance across different hardware architectures. Hardware specific optimization remains necessary to achieve efficient inference on the target architecture.

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With best wishes for your continued research and development.

Akira　　　　　　　　　　　　　　　　　　　　　Revised July 27, 2026